

Fourier transform of sinc function

Asked 10 years, 6 months ago Modified 11 months ago Viewed 126k times



Let us consider the Fourier transform of sinc function. As I know it is equal to a rectangular function in frequency domain and I want to get it myself, I know there is a lot of material about this, but I want to learn it by myself. We have sinc function which is defined as

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$$\text{sinc}(\omega_0 t) = \sin(\omega_0 t)/(\omega_0 t).$$



Its Fourier transform



$$\int_{\mathbb{R}} \sin(\omega_0 t) e^{-i\omega t}/(\omega_0 t) dt$$

can be represented as

$$\int_{\mathbb{R}} \sin(\omega_0 t) (\cos(\omega t) - i \sin(\omega t))/(\omega_0 t) dt.$$

Because we can distribute in brackets and consider that integral of difference is equal differences of integrals, we get

$$\int_{\mathbb{R}} \sin(\omega_0 t) \cos(\omega t)/(\omega_0 t) dt - \int_{\mathbb{R}} \sin(\omega_0 t) \sin(\omega t)/(\omega_0 t) dt$$

but first product is zero, right? Because sine and cosine are orthogonal, so how could continue? Please help me.

[calculus](#) [integration](#) [fourier-analysis](#) [definite-integrals](#) [indefinite-integrals](#)

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edited Feb 22, 2023 at 23:48

asked Apr 2, 2014 at 15:19



Mittens

41.6k

6

50

104



dato datuashvili

9,284

18

92

150



any help ?i could not continue – dato datuashvili Apr 2, 2014 at 18:54

4 Answers

Sorted by: Highest score (default)



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In terms of deriving the Fourier Transform, I will make some use of techniques highlighted in <http://www.clayturner.com/dsp/FTofSync.pdf>

Let us start with your expression

$$\int_{-\infty}^{\infty} \frac{\sin(\omega_0 t) \cos(\omega t)}{(\omega_0 t)} dt - \int_{-\infty}^{\infty} \frac{j \sin(\omega_0 t) \sin(\omega t)}{(\omega_0 t)} dt$$



Examining the integrals, the term $\omega_0 t$ in the denominator makes evaluating the integral more involved.



We can use some relevant trigonometric identities so that we can express



$$\sin(\omega_0 t) \cos(\omega t) = \frac{1}{2} [\sin((\omega + \omega_0)t) - \sin((\omega - \omega_0)t)]$$

$$\sin(\omega_0 t) \sin(\omega t) = \frac{1}{2} [\cos((\omega - \omega_0)t) - \cos((\omega + \omega_0)t)]$$

To deal with the awkward $\omega_0 t$ term, we can use the following identity to convert the single integral into a double integral that is far nicer to evaluate:

$$\frac{1}{\omega_0 t} = \int_0^{\infty} e^{-\omega_0 t s} ds$$

Thus the integral to evaluate is

$$\int_0^{\infty} \int_{-\infty}^{\infty} \sin(\omega_0 t) \cos(\omega t) e^{-\omega_0 t s} dt dx - j \int_0^{\infty} \int_{-\infty}^{\infty} \sin(\omega_0 t) \sin(\omega t) e^{-\omega_0 t s} dt dx$$

which expands to

$$\begin{aligned} & \frac{1}{2} \int_0^{\infty} \int_{-\infty}^{\infty} \sin((\omega + \omega_0)t) e^{-\omega_0 t s} dt dx - \frac{1}{2} \int_0^{\infty} \int_{-\infty}^{\infty} \sin((\omega - \omega_0)t) e^{-\omega_0 t s} dt dx \\ & - \frac{j}{2} \int_0^{\infty} \int_{-\infty}^{\infty} \cos((\omega - \omega_0)t) e^{-\omega_0 t s} dt dx + \frac{j}{2} \int_0^{\infty} \int_{-\infty}^{\infty} \cos((\omega + \omega_0)t) e^{-\omega_0 t s} dt dx \end{aligned}$$

We can exploit Fubini's theorem to rewrite the integral as

$$\begin{aligned} & \frac{1}{2} \int_{-\infty}^{\infty} \left[\int_0^{\infty} \sin((\omega + \omega_0)t) e^{-\omega_0 t s} dt \right] dx - \frac{1}{2} \int_{-\infty}^{\infty} \left[\int_0^{\infty} \sin((\omega - \omega_0)t) e^{-\omega_0 t s} dt \right] dx \\ & - \frac{j}{2} \int_{-\infty}^{\infty} \left[\int_0^{\infty} \cos((\omega - \omega_0)t) e^{-\omega_0 t s} dt \right] dx + \frac{j}{2} \int_{-\infty}^{\infty} \left[\int_0^{\infty} \cos((\omega + \omega_0)t) e^{-\omega_0 t s} dt \right] dx \end{aligned}$$

We shall use the following integral identities to calculate the above integrals :-

$$\int_0^{\infty} \sin(at) e^{-st} dt = \frac{a}{a^2 + s^2}$$

$$\int_0^{\infty} \cos(at) e^{-st} dt = \frac{s}{a^2 + s^2}$$

This results in

$$\frac{1}{2} \int_{-\infty}^{\infty} \left[\frac{(\omega + \omega_0)}{(\omega + \omega_0)^2 + \omega_0^2 x^2} \right] dx - \frac{1}{2} \int_{-\infty}^{\infty} \left[\frac{(\omega - \omega_0)}{(\omega - \omega_0)^2 + \omega_0^2 x^2} \right] dx$$

$$- \frac{j}{2} \int_{-\infty}^{\infty} \left[\frac{\omega_0 x}{(\omega - \omega_0)^2 + \omega_0^2 x^2} \right] dx + \frac{j}{2} \int_{-\infty}^{\infty} \left[\frac{\omega_0 x}{(\omega + \omega_0)^2 + \omega_0^2 x^2} \right] dx \quad (\text{Eq. 1})$$

To evaluate the real component integrals in (Eq. 1), we use the following result:-

$$\int_{-\infty}^{\infty} \frac{a}{a^2 + s^2} ds = \frac{|a|}{a} \int_{-\infty}^{\infty} \frac{1}{1 + y^2} dy = \text{sgn}(a) [\arctan y]_{-\infty}^{\infty} = \text{sgn}(a) \pi$$

This leads to

$$\int_{-\infty}^{\infty} \left[\frac{(\omega + \omega_0)}{(\omega + \omega_0)^2 + \omega_0^2 x^2} \right] dx = \text{sgn}(\omega + \omega_0) \pi$$

$$\int_{-\infty}^{\infty} \left[\frac{(\omega - \omega_0)}{(\omega - \omega_0)^2 + \omega_0^2 x^2} \right] dx = \text{sgn}(\omega - \omega_0) \pi$$

As regards the imaginary component integrals, note that the numerator is simply a constant times the derivative of the denominator, so we have

$$\int_{-\infty}^{\infty} \left[\frac{\omega_0 x}{(\omega - \omega_0)^2 + \omega_0^2 x^2} \right] dx = \frac{1}{2\omega_0} [\ln\{(\omega - \omega_0)^2 + \omega_0^2 x^2\}]_{-\infty}^{\infty}$$

$$\int_{-\infty}^{\infty} \left[\frac{\omega_0 x}{(\omega + \omega_0)^2 + \omega_0^2 x^2} \right] dx = \frac{1}{2\omega_0} [\ln\{(\omega + \omega_0)^2 + \omega_0^2 x^2\}]_{-\infty}^{\infty}$$

Combining the imaginary component integrals as per the integral in (Eq. 1) we wish to evaluate and noting that they have opposite signs, we have

$$- \int_{-\infty}^{\infty} \left[\frac{\omega_0 x}{(\omega - \omega_0)^2 + \omega_0^2 x^2} \right] dx + \int_{-\infty}^{\infty} \left[\frac{\omega_0 x}{(\omega + \omega_0)^2 + \omega_0^2 x^2} \right] dx$$

$$= \lim_{x \rightarrow \infty} \frac{1}{2\omega_0} \ln \left[\frac{(\omega + \omega_0)^2 + \omega_0^2 x^2}{(\omega - \omega_0)^2 + \omega_0^2 x^2} \right] + \lim_{x \rightarrow -\infty} \frac{1}{2\omega_0} \ln \left[\frac{(\omega - \omega_0)^2 + \omega_0^2 x^2}{(\omega + \omega_0)^2 + \omega_0^2 x^2} \right] = \frac{2}{\omega_0} \ln(1) = 0$$

Thus the imaginary terms in the integral cancel out, leading to the integral in (Eq. 1) being a real result, as follows

$$\frac{\pi}{2} [\text{sgn}(\omega + \omega_0) - \text{sgn}(\omega - \omega_0)]$$

Putting everything together we have:

$$\int_{-\infty}^{\infty} \frac{\sin(\omega_0 t) \cos(\omega t)}{(\omega_0 t)} dt - \int_{-\infty}^{\infty} \frac{j \sin(\omega_0 t) \sin(\omega t)}{(\omega_0 t)} dt = \frac{\pi}{2} [\text{sgn}(\omega + \omega_0) - \text{sgn}(\omega - \omega_0)]$$

The result is a rectangular function that starts from frequency $-\omega_0$ and ends at frequency ω_0 .

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edited Apr 9, 2014 at 6:16

answered Apr 8, 2014 at 17:34

 Alijah Ahmed
11.7k 2 21 22

 You are welcome - the website I've linked in the answer proved a great resource. – Alijah Ahmed Apr 8, 2014 at 17:40

 a bit deep, but i get point – dato datuashvili Apr 8, 2014 at 17:40

 I agree it's quite involved, but I thought deriving the result from first principles would be an interesting exercise. – Alijah Ahmed Apr 8, 2014 at 17:41

 interesting is Fubini's theorem by the way – dato datuashvili Apr 8, 2014 at 17:42

 what does it says if you could explain? – dato datuashvili Apr 8, 2014 at 17:43

|



$$f(t) \equiv \frac{\sin(\omega_0 t)}{\omega_0 t} = \int_{-\infty}^{\infty} \hat{f}(\omega) e^{-i\omega t} \frac{d\omega}{2\pi}, \quad \hat{f}(\omega) : ?$$

11



$$\begin{aligned}\hat{f}(\omega) &= \int_{-\infty}^{\infty} \operatorname{sinc}(\omega_0 t) e^{i\omega t} dt = \frac{1}{|\omega_0|} \int_{-\infty}^{\infty} \frac{\sin(t)}{t} e^{i\omega t/|\omega_0|} dt \\ &= \frac{1}{|\omega_0|} \int_{-\infty}^{\infty} \left(\frac{1}{2} \int_{-1}^1 e^{i\nu t} d\nu \right) e^{i\omega t/|\omega_0|} dt = \frac{1}{2|\omega_0|} \int_{-1}^1 d\nu \int_{-\infty}^{\infty} \overbrace{e^{i(\nu + \omega/|\omega_0|)t}}^{2\pi \delta\left(\nu + \frac{\omega}{|\omega_0|}\right)} dt\end{aligned}$$

where $\delta(x)$ is the [Dirac Delta 'Function'](#).

Then

$$\hat{f}(\omega) = \frac{\pi}{|\omega_0|} \Theta\left(1 - \left|\frac{\omega}{|\omega_0|}\right|\right)$$

where $\Theta(x)$ is the [Heaviside Step Function](#).

$$\hat{f}(\omega) = \frac{\pi}{|\omega_0|} \Theta(|\omega_0| - |\omega|)$$

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edited Nov 30, 2016 at 0:14

answered Apr 9, 2014 at 7:24



Felix Marin

91.7k 10 173 206

▲ @datodatuashvili You're welcome. I enjoyed it. – Felix Marin Apr 9, 2014 at 7:58

7 This simple method is missing. Since for $\omega_0 \geq 0$,

$$\begin{aligned}\int_{\mathbb{R}} \mathbf{1}_{[-\omega_0, \omega_0]}(\omega) e^{i\omega t} d\omega &= \int_{-\omega_0}^{\omega_0} e^{i\omega t} d\omega \\ &= \frac{e^{i\omega_0 t} - e^{-i\omega_0 t}}{it} \\ &= 2\omega_0 \operatorname{sinc}(\omega_0 t)\end{aligned}$$

one deduces by the Fourier inversion theorem that

$$\int_{\mathbb{R}} \operatorname{sinc}(\omega_0 t) e^{-i\omega t} dt = \frac{\pi}{\omega_0} \mathbf{1}_{[-\omega_0, \omega_0]}(\omega).$$

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edited Oct 31, 2023 at 8:06

answered Feb 22, 2023 at 23:32



LL 3.14

13.1k 2 16 48

5 We know that the Fourier transform of $\operatorname{Sinc}(z)$ is,

$$\int_{-\infty}^{+\infty} \frac{\sin(z)}{z} e^{-i\omega z} dz$$

and

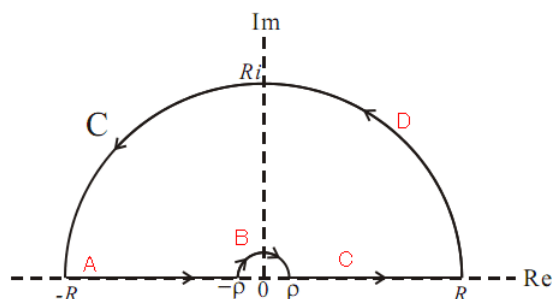
$$\int_{-\infty}^{+\infty} \frac{\sin(z)}{z} e^{-i\omega z} dz = \int_{-\infty}^{+\infty} \frac{e^{iz} - e^{-iz}}{2iz} e^{-i\omega z} dz.$$

So,

(1)

$$\int_{-\infty}^{+\infty} \frac{\sin(z)}{z} e^{-i\omega z} dz = \frac{1}{2i} \int_{-\infty}^{+\infty} \frac{e^{i(1-\omega)z}}{z} dz + \frac{1}{2i} \int_{+\infty}^{-\infty} \frac{e^{-i(1+\omega)z}}{z} dz.$$

Let us consider the first item, when $\omega < 1$, namely $(1-\omega) > 0$, we can choose the path below to do the contour integration.



Using the method of complex residues, we take the contour with no singular point, separating the path into four parts, namely A, B, C and D shown as the red letters in the figure. Thus,

(2)

$$\oint \frac{e^{i(1-\omega)z}}{z} dz = \lim_{\rho \rightarrow 0} \int_{-R}^{-\rho} \frac{e^{i(1-\omega)z}}{z} dz + \lim_{\rho \rightarrow 0} \int_{\rho}^R \frac{e^{i(1-\omega)z}}{z} dz + \lim_{R \rightarrow \infty} \int_{\rho}^R \frac{e^{i(1-\omega)z}}{z} dz + \lim_{R \rightarrow \infty} \int_D \frac{e^{i(1-\omega)z}}{z} dz = 2\pi i \operatorname{Res} f(z) = 0.$$

Also if $(1-\omega) > 0$, namely $\omega < 1$, and when $R \rightarrow \infty$ and $\rho \rightarrow 0$, it is easy to prove using Jordan's lemma, that the fourth term is equal to zero. Thus we can rewrite the rest items as,

$$\int_{-\infty}^0 \frac{e^{i(1-\omega)z}}{z} dz + \int_0^{+\infty} \frac{e^{i(1-\omega)z}}{z} dz = \int_{-\infty}^{+\infty} \frac{e^{i(1-\omega)z}}{z} dz = -\lim_{\rho \rightarrow 0} \int_B \frac{e^{i(1-\omega)z}}{z} dz.$$

In order to calculate the value of the item on the right side, I use $z = \rho e^{i\theta}$, so $dz = i\rho e^{i\theta} d\theta$, then we have,

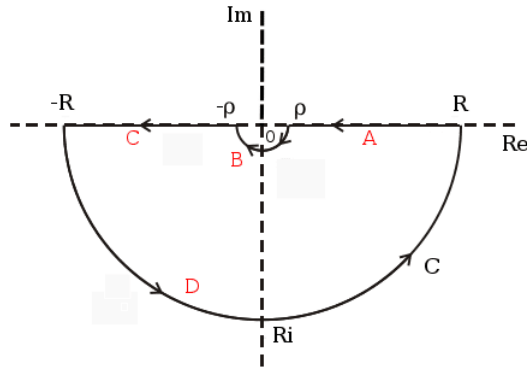
$$\lim_{\rho \rightarrow 0} \int_B \frac{e^{i(1-\omega)z}}{z} dz = \lim_{\rho \rightarrow 0} \int_{\pi}^0 \exp[i(1-\omega)\rho e^{i\theta}] i d\theta = \int_{\pi}^0 i d\theta = i(0 - \pi) = -i\pi.$$

So, in the end, we get the value of the first integration item shown in the Eq.(1),

(3)

$$\int_{-\infty}^{+\infty} \frac{e^{i(1-\omega)z}}{z} dz = i\pi, \quad \omega < 1.$$

When $\omega > 1$, according to Jordan's lemma, we can not get the fourth item in the Eq.(2) to be zero. So we have to choose another path for finalising the integration. So I chose the path below to deal with such a situation,



It is not hard to calculate the same way as the previous one,

(4)

$$\oint \frac{e^{i(1-\omega)z}}{z} dz = \lim_{\rho \rightarrow 0} \int_R^\rho \frac{e^{i(1-\omega)z}}{z} dz + \lim_{\rho \rightarrow 0} \int_B \frac{e^{i(1-\omega)z}}{z} dz + \lim_{R \rightarrow \infty} \int_{-\rho}^{-R} \frac{e^{i(1-\omega)z}}{z} dz + \lim_{R \rightarrow \infty} \int_D \frac{e^{i(1-\omega)z}}{z} dz = 2\pi i \operatorname{Res} f(z) = 0,$$

and here again I use the same trick, $z = R e^{i\theta}$, so $dz = i R e^{i\theta} d\theta$. Because $\omega > 1$,

$$\lim_{R \rightarrow \infty} \int_B \frac{e^{i(1-\omega)z}}{z} dz = \lim_{R \rightarrow \infty} \int_{\pi}^0 \exp[i(1-\omega)R e^{i\theta}] i d\theta = \int_{\pi}^0 0 \times i d\theta = 0,$$

$$\lim_{\rho \rightarrow 0} \int_B \frac{e^{i(1-\omega)z}}{z} dz = \lim_{\rho \rightarrow 0} \int_0^{\pi} \exp[i(1-\omega)\rho e^{i\theta}] i d\theta = \int_0^{\pi} i d\theta = i(\pi - 0) = i\pi.$$

So, in the end,

(5)

$$\int_{-\infty}^{+\infty} \frac{e^{i(1-\omega)z}}{z} dz = i\pi, \quad \omega > 1.$$

Now, let's move on to the second part in the Eq.(1). Following the same steps, using contour above real axis I get,

(6)

$$\int_{+\infty}^{-\infty} \frac{e^{-i(1+\omega)z}}{z} dz = \lim_{\rho \rightarrow 0} \int_{\pi}^0 \exp[-i(1+\omega)\rho e^{i\theta}] i d\theta = -i\pi, \quad \omega < -1;$$

and using contour below real axis I get,

(7)

$$\int_{+\infty}^{-\infty} \frac{e^{-i(1+\omega)z}}{z} dz = \lim_{\rho \rightarrow 0} \int_0^{\pi} \exp[-i(1+\omega)\rho e^{i\theta}] i d\theta = i\pi, \quad \omega > -1.$$

Finally, we can write the answer as, if $-1 < \omega < 1$

$$\int_{-\infty}^{+\infty} \frac{\sin(z)}{z} e^{-i\omega z} dz = \frac{1}{2i} \int_{-\infty}^{+\infty} \frac{e^{i(1-\omega)z}}{z} dz + \frac{1}{2i} \int_{+\infty}^{-\infty} \frac{e^{-i(1+\omega)z}}{z} dz = \frac{\pi}{2} + \frac{\pi}{2} = \pi;$$

and if $\omega > 1$ and $\omega < -1$

$$\int_{-\infty}^{+\infty} \frac{\sin(z)}{z} e^{-i\omega z} dz = \frac{1}{2i} \int_{-\infty}^{+\infty} \frac{e^{i(1-\omega)z}}{z} dz + \frac{1}{2i} \int_{+\infty}^{-\infty} \frac{e^{-i(1+\omega)z}}{z} dz = \frac{\pi}{2} - \frac{\pi}{2} = 0.$$

(Note: if we define the Fourier Transform as

$$\int_{-\infty}^{+\infty} \frac{\sin(z)}{z} e^{-2\pi i \omega z} dz$$

, Then the result could be a little different.)

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answered Jun 7, 2018 at 19:28



CyTex

89 1 3



Note that $\lim_{R \rightarrow \infty} \exp[i(1-\omega)Re^{i\theta}] = 0$ for $\omega > 1$ is not true. – S.H.W Jun 8, 2020 at 23:55